3.14.2 Internal voltage reference (V_{REFINT})

The internal voltage reference (V_{REFINT}) provides a stable (bandgap) voltage output for the ADC. V_{REFINT} is internally connected to an ADC input. The V_{REFINT} voltage is individually precisely measured for each part by ST during production test and stored in the part's engineering bytes. It is accessible in read-only mode.

Table 6. Internal voltage reference calibration values

Calibration value name	Description	Memory address
V_{REFINT}	Raw data acquired at a temperature of 30 °C (± 5 °C), V_{DDA} = V_{REF+} = 3.0 V (± 10 mV)	0x1FFF756A-0x1FFF756B

3.15 Timers and watchdogs

The device includes an advanced-control timer, five general-purpose timers, two watchdog timers and a SysTick timer. *Table 7* compares features of the advanced-control and general-purpose timers.

Table 7. Timer feature comparison

Timer	Timer type	Counter resolution	Counter type	Maximum operating frequency	Prescaler factor	DMA request generation	Capture/ compare channels	Comple-mentary outputs
TIM1	Advanced-control	16-bit	Up, down, up/down	48 MHz	Integer from 1 to 2^{16}	Yes	4 +2 internal	3
TIM2	General-purpose	32-bit	Up, down, up/down	48 MHz	Integer from 1 to 2^{16}	Yes	4	-
TIM3	General-purpose	16-bit	Up, down, up/down	48 MHz	Integer from 1 to 2^{16}	Yes	4	-
TIM14	General-purpose	16-bit	Up	48 MHz	Integer from 1 to 2^{16}	No	1	-
TIM16 TIM17	General-purpose	16-bit	Up	48 MHz	Integer from 1 to 2^{16}	Yes	1	1

3.15.1 Advanced-control timer (TIM1)

The advanced-control timer can be seen as a three-phase PWM unit multiplexed on 6 channels. It has complementary PWM outputs with programmable inserted dead-times. It can also be seen as a complete general-purpose timer. The four independent channels can be used for:

- input capture
- output compare
- PWM output (edge or center-aligned modes) with full modulation capability (0-100%)
- one-pulse mode output

On top of these, there are two internal channels that can be used.